

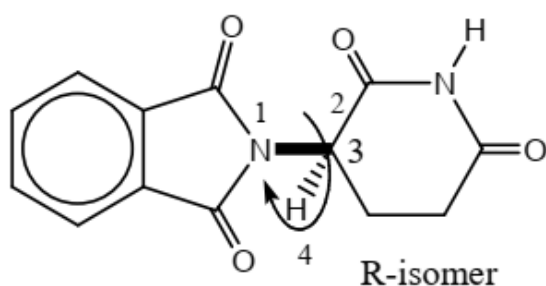
Stereochemistry:

It involves the study of the relative spatial arrangement of atoms within the molecules.

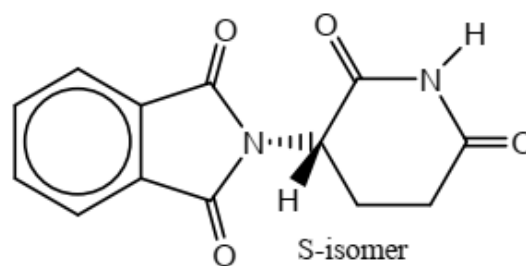
Dynamic stereochemistry: Dynamic stereochemistry is the study of the effect of stereochemistry on the rate of a chemical reaction.

First Stereochemist – Louis Pasteur (1849):

Significance of stereochemistry: One of the most infamous demonstration of the significance of stereochemistry was the **Thalidomide disaster**. Thalidomide is a drug was first prepared in 1957 in Germany, prescribed for treating morning Sickness in pregnant women. It was discovered that one optical isomer i.e. RIsomer of the drug was safe whereas the S-isomer had teratogenic effect, causing serious genetic damage to early embryonic growth and development.



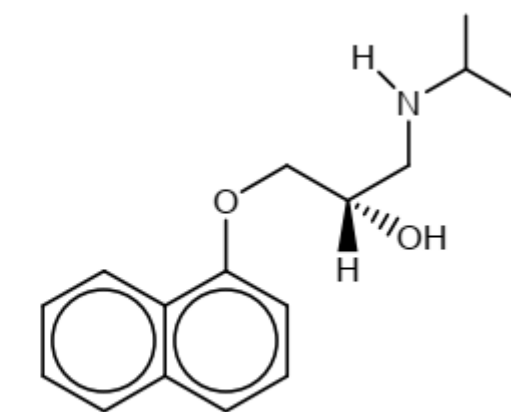
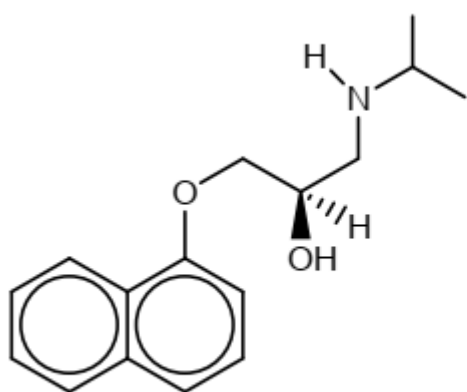
Drug for morning sickness in pregnant women.



Teratogenic effect

[**Remark:** In human body, Thalidomide undergoes racemization: even if only one of the two stereoisomers is ingested, the other one is produced.]

Now we have another example - Propanolol.



SOME TERMINOLOGY

Optical activity: The term optical activity derived from the interaction of chiral materials with polarized light.

Scalemic: Any non-racemic chiral substance is called *Scalemic*.

- A chiral substance is enantio pure or homochiral when only one of two possible enantiomer is present.
- A chiral substance is enantio enriched or heterochiral when an excess of one enantiomer is present but not the exclusion of the other.

Three terms are used to designate a carbon atom bonded tetrahedrally to four different substituents in a chiral molecule.

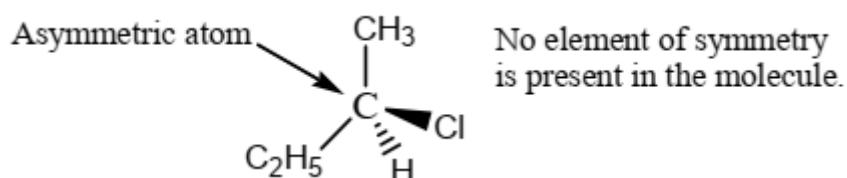
(a) Asymmetric atom (*LeBell & Vant Hoff* for an atom attached with 4 different groups).

(b) Chiral centre

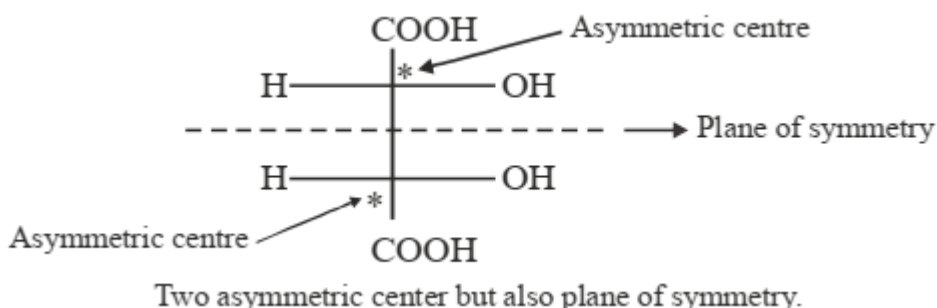
(c) Stereocentre.

Asymmetric atom:

Compounds with one such atom are truly asymmetric as they lack symmetry. For example



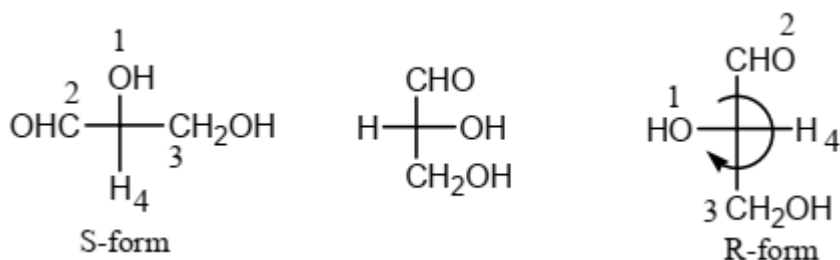
However, there are molecules which also have atoms with four different substituents and which also have various symmetry element including plane of symmetry as in mesotartaric acid.



Chirality is a geometric property which influences and affects all parts of a chiral molecule.

Stereogenic centre or Stereocentre:

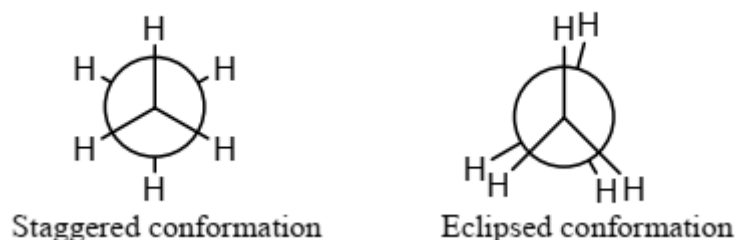
A stereogenic centre or in short a stereocentre is an atom having groups of such nature that an interchange of any two groups will produce a stereocentre.



A carbon atom that is a stereocentre is also called a stereogenic carbon.

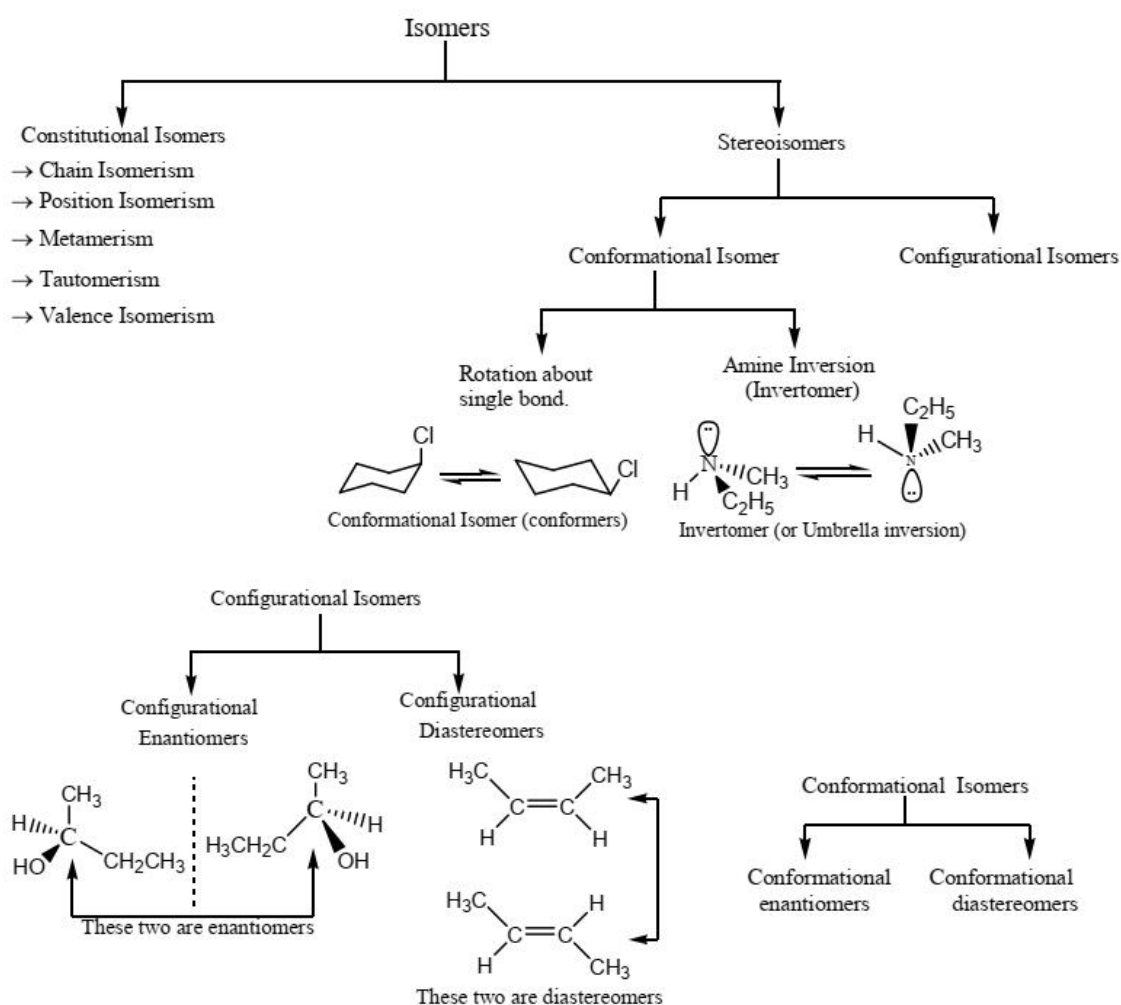
Conformation: Structures that can be interconverted simply by rotation about single bonds are conformation of the same molecule.

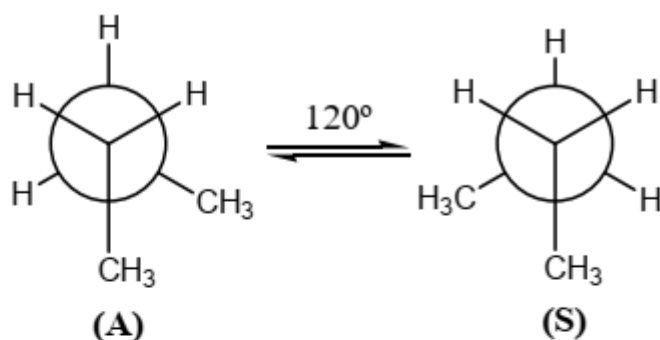
For example:



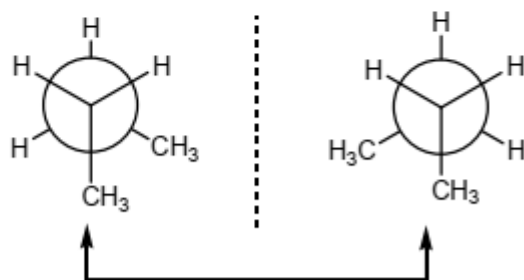
Note: These two are the conformation of ethane arises due to rotational possibilities across C-C single bond.

Configuration: Structures that can be interconverted only by breaking one or more bonds have different configuration and they are stereoisomers specifically known as configurational isomers.

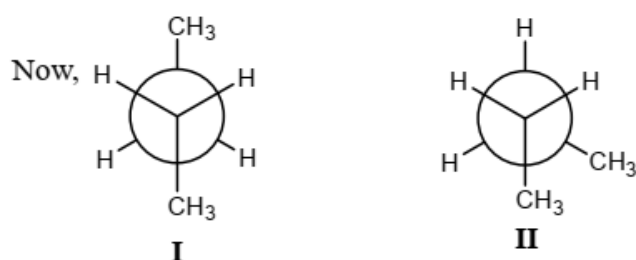




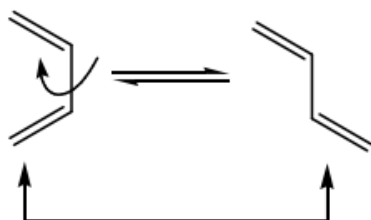
So, (A) and (S) are mirror image of each other as shown below



So, these two are conformational enantiomers



Since, I and II are not mirror image to each other so these two are conformational diastereomers.

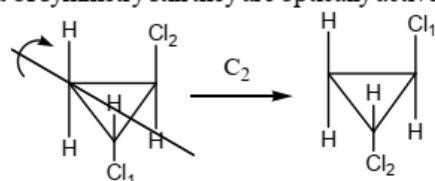


Conformational diastereomers

Essential criteria for a molecule to be chiral. There is no any single criterion.

1. There must be lack of element of symmetry.

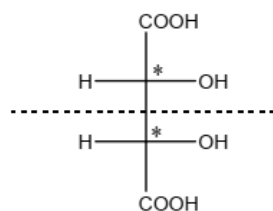
Note: It is not necessary and sufficient condition because there are some set of molecules which have some element of symmetry still they are optically active. For example.



Optically active but having C_2 -symmetry.

2. The carbon in the molecule should be attached to four different groups.
It is not a necessary and sufficient condition also because we have an example in which carbon have four different groups but it is still optically inactive.

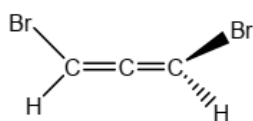
For example:



Two asymmetric centre but still optically inactive owing to plane of symmetry.

On the other hand we have also an example in which there is not any chiral centre but still molecule is optically active.

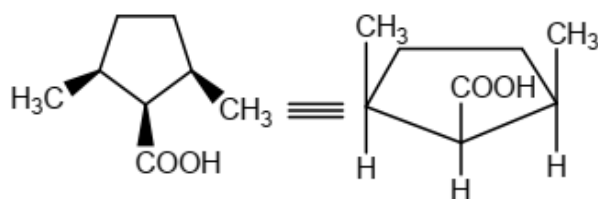
For example: Properly substituted allene.



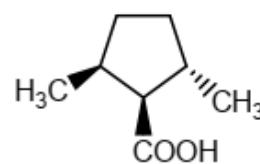
Not any chiral centre but still it is optically active.

Remrak: This compound is optically active not due to chiral centre but due to chiral axis.

3. There should be an absence of plane of symmetry.

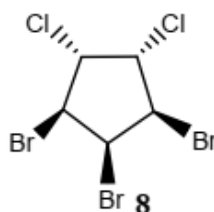
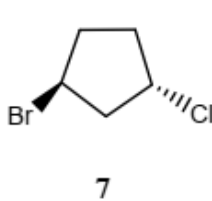
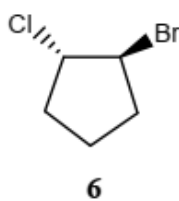
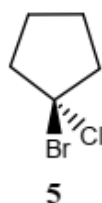
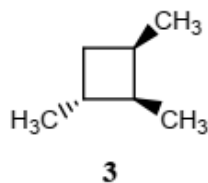
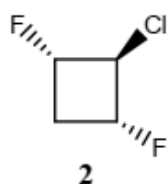


→ Plane of symmetry
→ Achiral molecule
→ Optically inactive.



→ No plane of symmetry
→ Chiral molecule
→ Optically active

Problem-2: Find out which molecules are/is optically active.



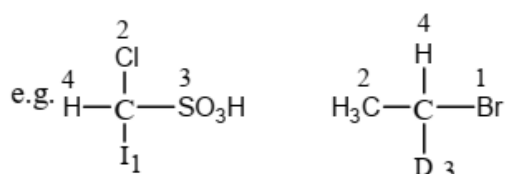
Specification of configuration R/S: Then is absolute configuration of chiral centre
Proposed by R.S. Chan, Sir Christopher Ingold, V. Prelog.

Sequence rule: Priority to the four atoms or groups of atoms attached to the chiral centre can be determined in accordance with sequence rule which are as follows.

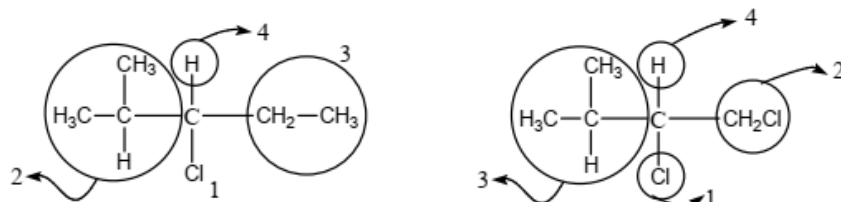
Rule 1: If the four atoms attached to the chiral centre are all different, priority depends on atomic number, with the atom of high atomic number getting higher priority.

Rule 2: In case of isotope, the atom of higher mass number has the higher priority.

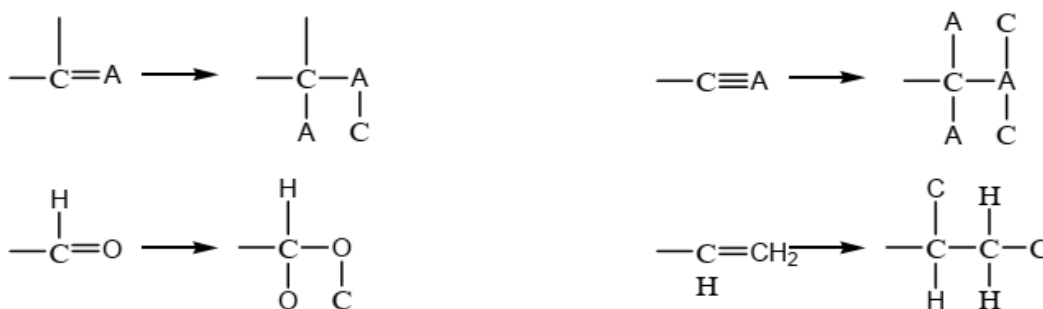
For example, among Br, C, D, H priority order is $\text{Br} > \text{C} > \text{D} > \text{H}$.



Rule 3: If the relative priority of two groups cannot be decided by rule mentioned above, then look for next atoms.

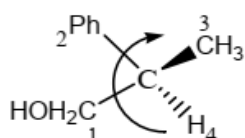
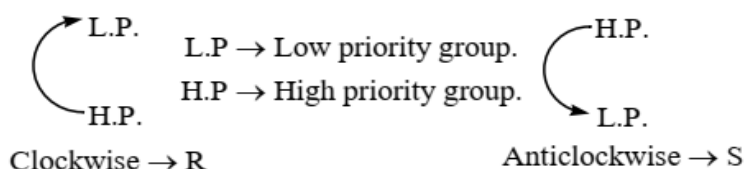


Rule 4: Where there is a double bond or triple bond, both atoms are considered to be duplicated or triplicated.

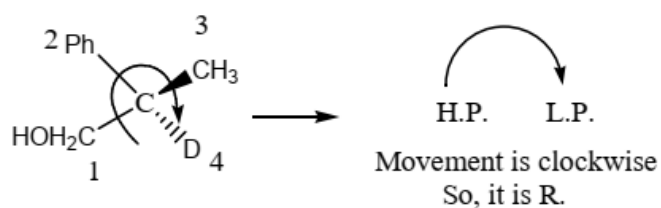


- (1) Assign the priority sequence by above mentioned method.
- (2) Find out position of the 4th group.
- (3) Connect 1→2→3 making a circle.

Case I: If fourth group is below the plane.

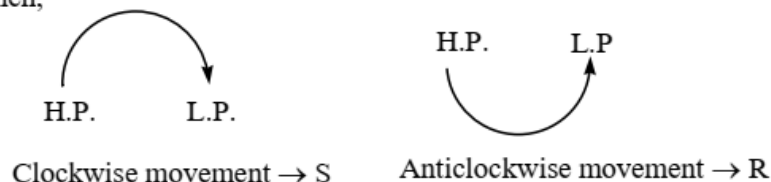


Note: In this case 4th group 'H' is below the plane and rotation is clockwise so, it is 'R'.

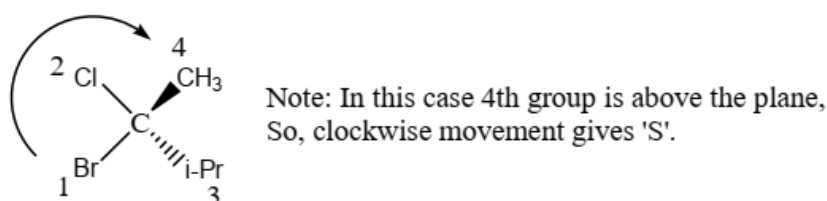


Case 2: If fourth group is above the plane.

Then,

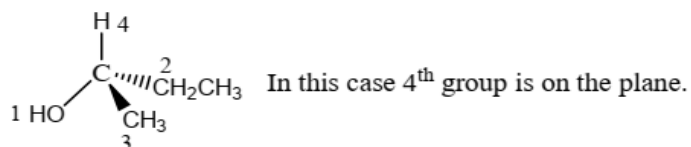


For example:

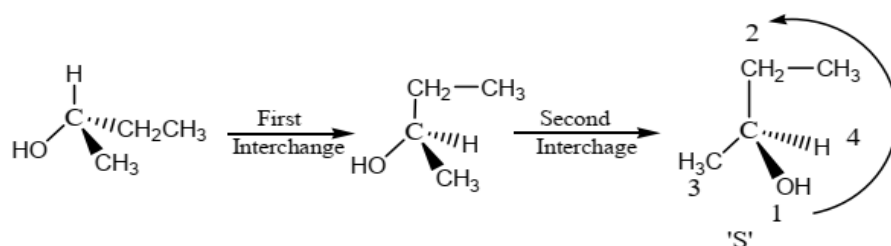


Case III: If the fourth group is on the plane then do double interchange in such a way that the 4th group undergoes below the plane.

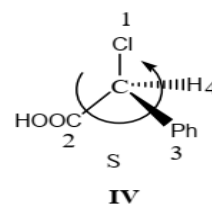
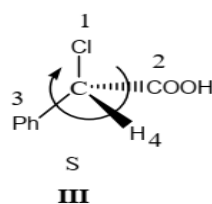
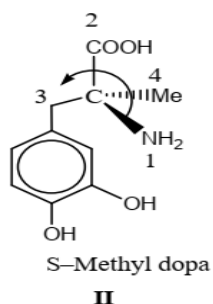
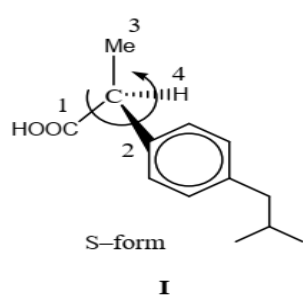
For example:

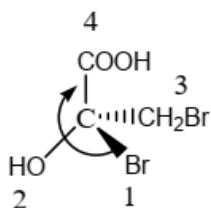


Double interchange can be done as

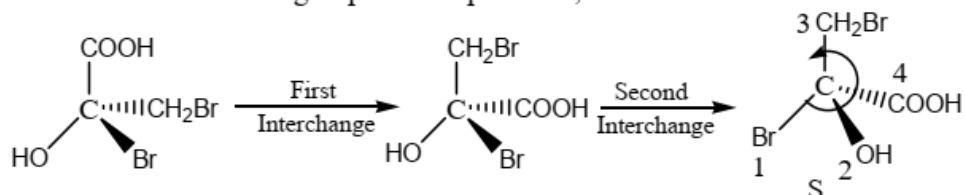


SOLVED PROBLEMS

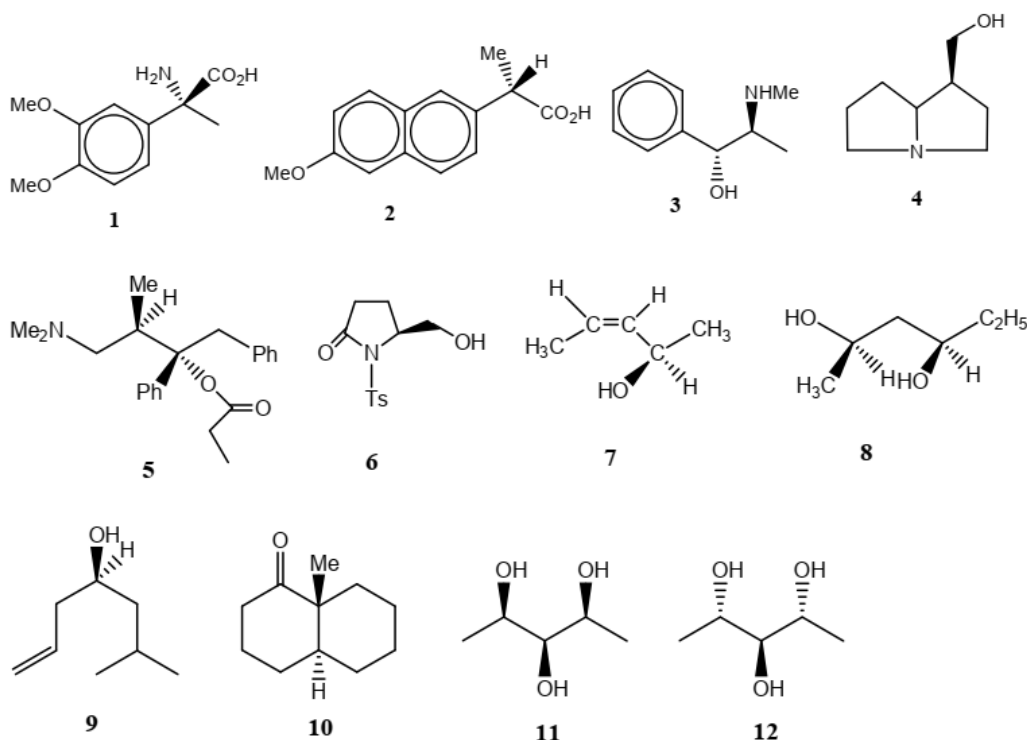




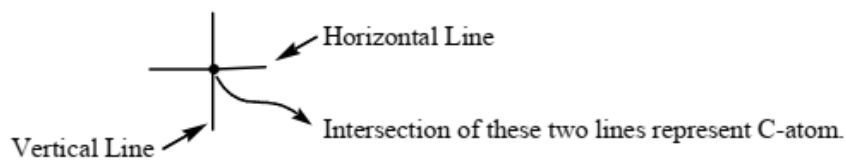
Note: In this case fourth group is on the plane. So, we will have to do double interchange as shown below



1. Find R/S of the following compounds

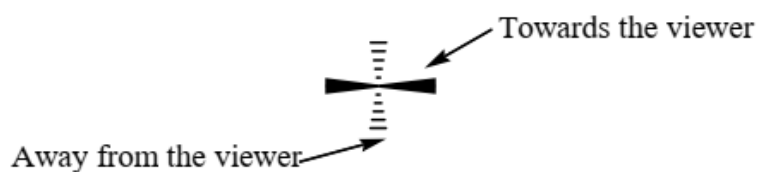


R/S Nomenclature in Fischer projection.



Vertical Line → away from the viewer.

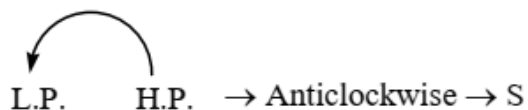
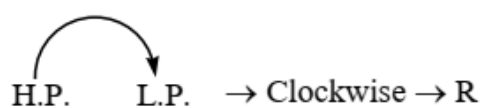
Horizontal Line → Towards the viewer.



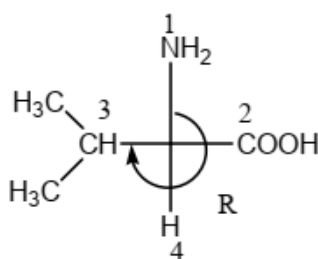
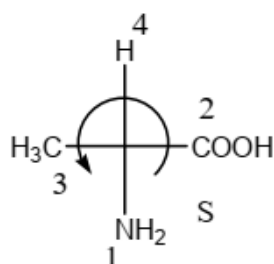
For R/S nomenclature.

Assign priority sequence.

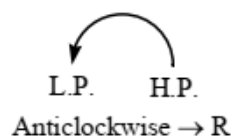
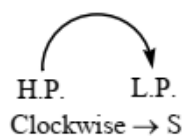
Case I: If fourth group is present on the top or bottom of vertical line then



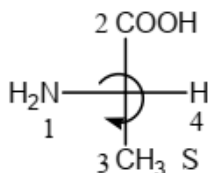
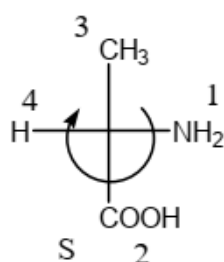
For example:



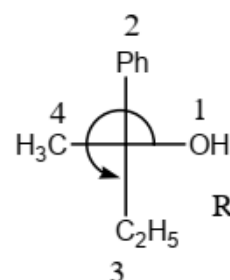
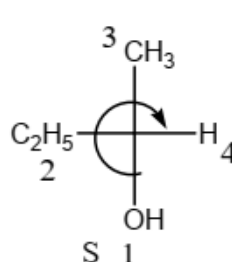
Case II: If 4th group is present on left or right side of the horizontal line then.



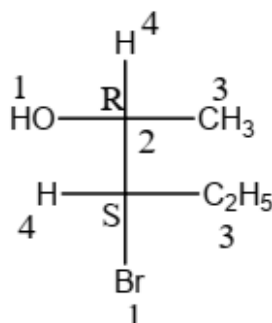
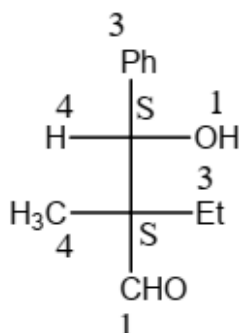
For example:



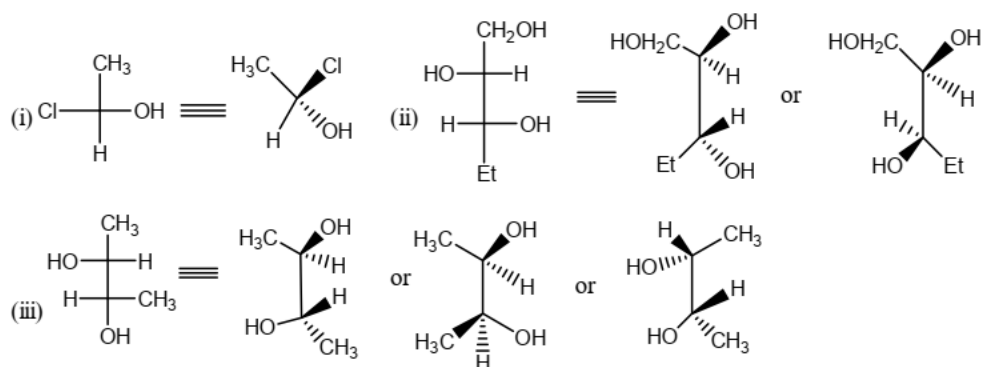
;



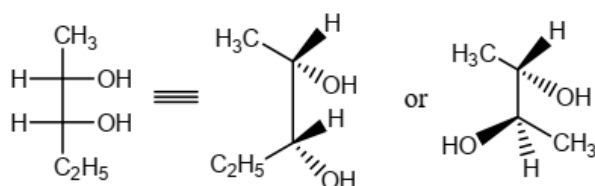
For example:



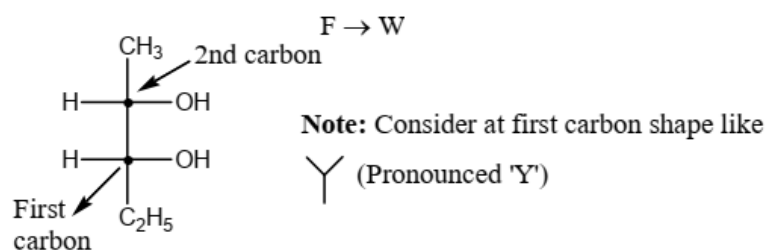
Conversion of Fisher → Wedge.



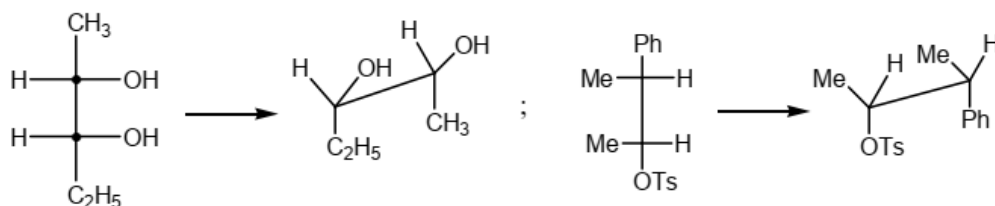
Note: In all conversion, the configuration (R/S) should not be change.



Conversion of fischer to sawhorse.

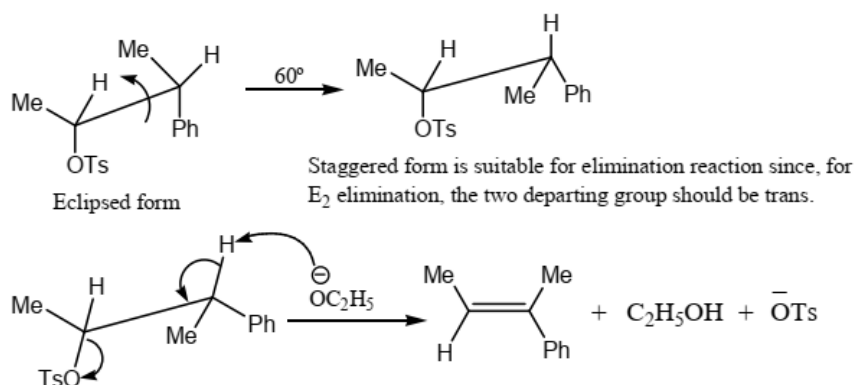


and then combine these two by a straight line like

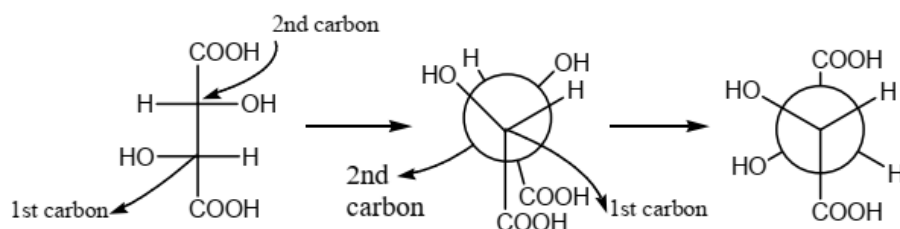
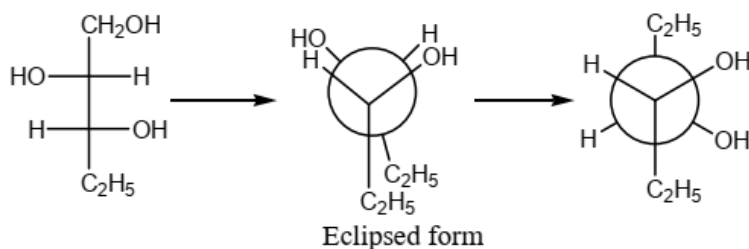
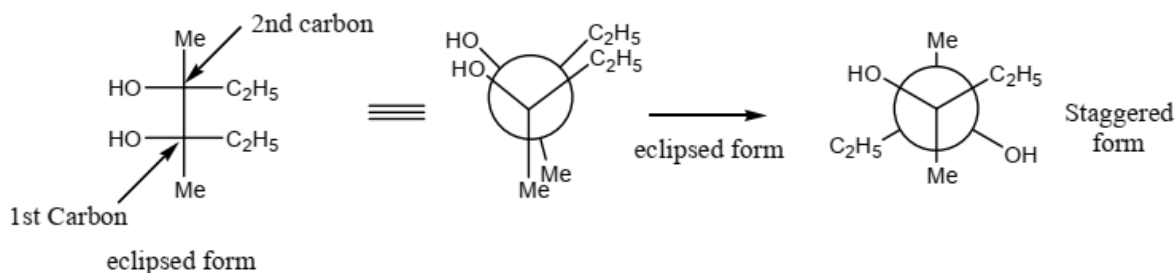


Note: Since Fischer projection is represented in eclipsed form so the resulting sawhorse should also be in eclipsed form if we have need of staggered form we can obtain it by simple rotation.

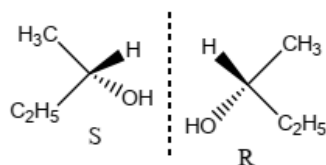
For example:



Conversion of Fischer to Newmann:

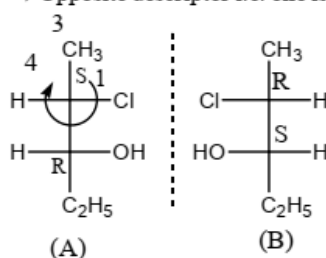


Enantiomers: Enantiomers are the stereoisomers which are non superimposable mirror images to each other. So these two stereoisomers have opposite descriptor.

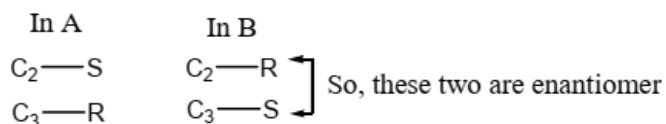


→ Non super imposable mirror image.

→ Opposite descriptor i.e. one is R and other is S.



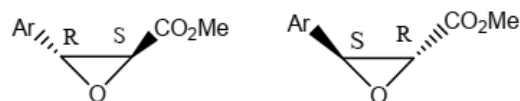
So, in compound 'A' and 'B', the configuration at chiral centre are



Properties of enantiomer:

1. All physical properties such as M.P., refractive index, vapour pressure, relative density, NMR spectrum, IR spectrum are same except direction of optical rotation. (Magnitude is same but direction is opposite).
2. All the chemical properties of enantiomers towards achiral reagent are always identical.
3. The chemical properties of enantiomers will be different in the following condition.

- Note:** If we run NMR spectrum of enantiomers in chiral solvent then it will be also different.
 3. What is the relation between following compound.



So, these two compounds are enantiomer.

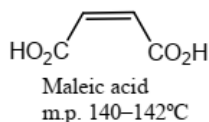
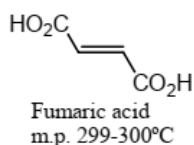
Diastereoisomers:

Diastereomers are the stereoisomers that are not enantiomers.

Some important points regarding the diastereoisomers.

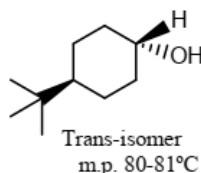
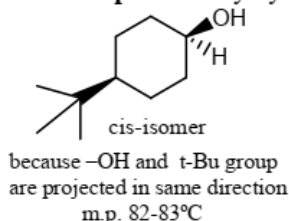
1. Diastereomers can arise when structures have more than one stereogenic centre.
2. The magnitude of optical rotation of diastereoisomers are always different but the direction may be same or opposite.
3. The physical properties of diastereomers are always different but difference may be more or less.
4. The chemical properties of diastereomers toward chiral as well as achiral reagent is always different.

Examples of diastereoisomers:



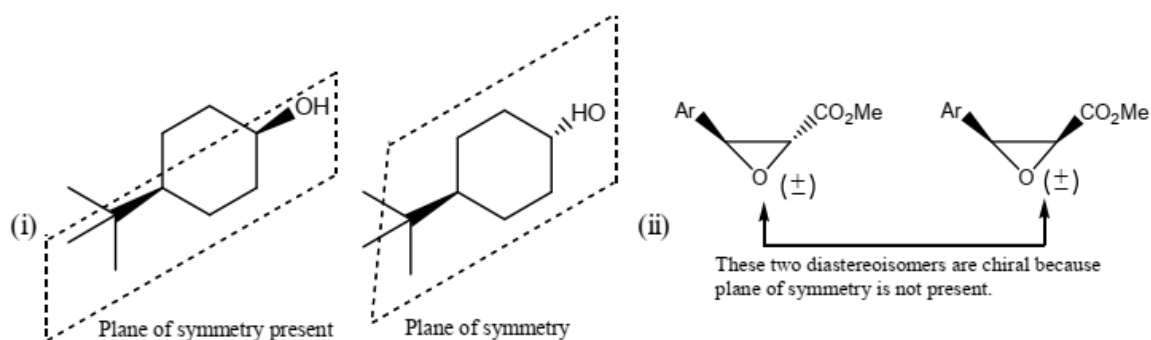
Note: geometrical isomers (i.e. cis and trans isomers) are always diastereoisomers. A similar stereoisomers can exist in cyclic compounds.

For example: 4-t-butyl cyclohexanol.

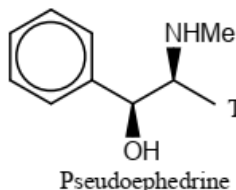
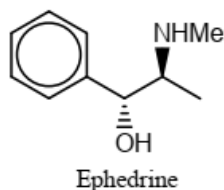


So, these two stereoisomers are called diastereomers.

Note: Diastereoisomers can be chiral or achiral.



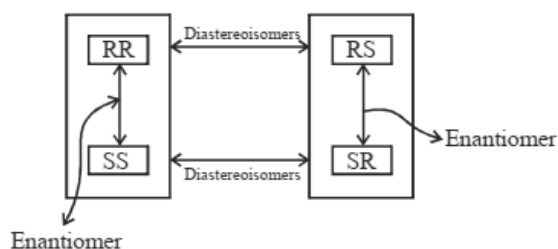
So, achiral diastereoisomers.



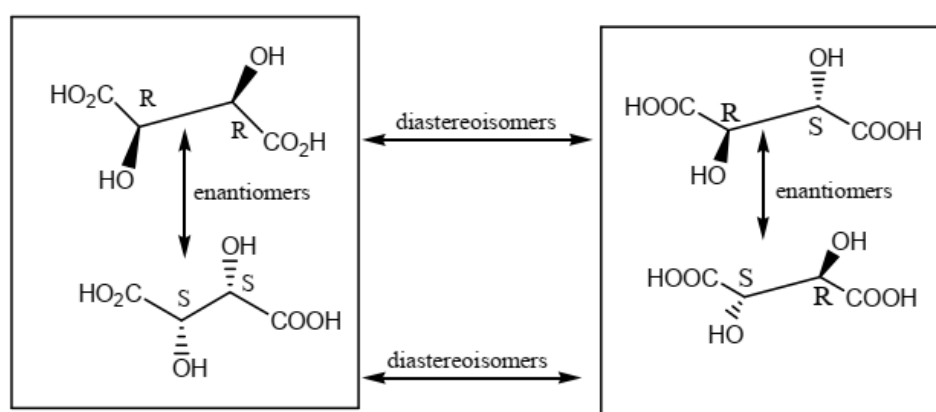
These two are diastereomers.

Remark: The diastereoisomers are different compounds with different names (For example ephedrine and pseudoephedrine) and different properties, while the pair of enantiomers are the same compound and differ only in the direction in which they rotate plane polarized light.

In summary if we have compound having two chiral centre each one R/S then



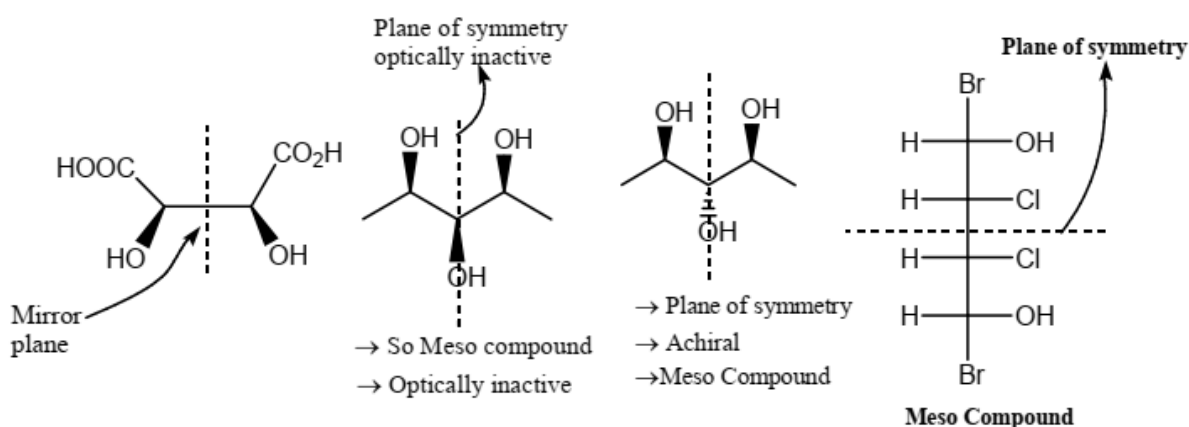
For example:



Meso Compounds:

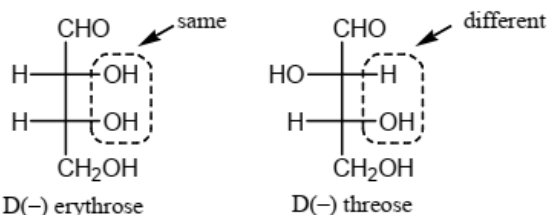
Compounds that contain stereogenic centres but achiral are called meso compounds. This means that there is a plane of symmetry with R stereochemistry on one side and 'S' stereochemistry on the other.

For example:



Threo and Erythro nomenclature

Threo and Erythro nomenclature are based on sugar chemistry.



- Erythrose and threose are diastereomers.
- A molecule with two adjacent stereocentres, when there are two groups which are common to each carbon while third is different i.e. $C_{abx} - C_{aby}$ gives rise to erythro and threo diastereomer.

Procedure for finding erythro and threo diastereomers.

Step I: Find out the group or atom which is not common on two adjacent asymmetric centre.

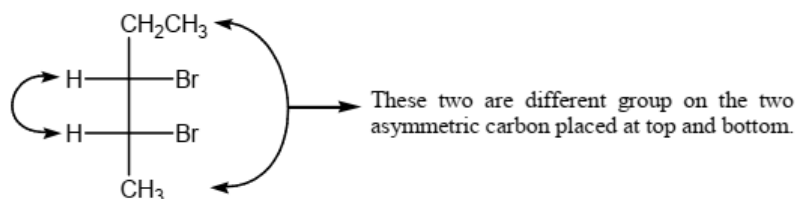
Step II: Placed this different group on top and bottom position of vertical line in Fischer projection formula.

Step III: And arrange the rest group or atoms around horizontal position of Fischer projection.

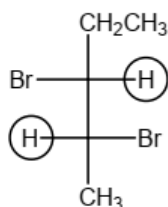
Case I: If two similar group are on the **same side** then it is called **erythro**.

Case II: If two similar group on the **opposite side** then it is called **threo**.

For example:



These two H₂ atoms are projected in same side so, it is erythro



In this structure the two similar group viz H and Br are opposite so, it is threo.

Note: The terms erythro and threo are generally applied only to those molecules which do not have symmetric ends.

In summary, condition for erythro and threo nomenclature:

- Two asymmetric carbon should be there
- On two asymmetric carbon, two of the groups are the same and the third is different.